

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 24
 - Riemann-Stieltjes integrals

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Riemann sums

A Riemann sum of $f(x)$ on $[a, b]$ with n is

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$$R(P, f, \{t_k\}) = \sum_{k=1}^n f(t_k) \Delta x_k$$

where t_k and $\Delta x_k := x_k - x_{k-1}$ come from:

- a **partition** $P = \{x_0 = a, x_1, \dots, x_n = b\}$
- a set of n **sample points** $t_k \in [x_{k-1}, x_k]$ with $1 \leq k \leq n$

Riemann integrability

Definition (Very informal first try)

The Riemann integral is the limit of *all* Riemann sums as we increase the number of rectangles.

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Question

What problems do you see with this definition?

Some issues

Problems with definition:

- for n rectangles, there are uncountably many partitions!
- for a specific partition, there are uncountably many sample points!

How do we do a limit in a place with so many choices?!

Coarse vs fine

Definition

If P and P' are two partitions of $[a, b]$ with $P \subseteq P'$, we say P is **coarser** than P' and P' is **finer** than P .
Any finer partition is called a **refinement**.

Coarse vs fine

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Examples:

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Examples:

- $\{1, 2, 3\}$ is **finer** than $\{1, 3\}$

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Examples:

- $\{1, 2, 3\}$ is **finer** than $\{1, 3\}$
- $\{1, 1.1, 2, 2.5, 3\}$ is **coarser** than $\{1, 1.1, 1.5, 2, 2.5, 2.9, 3\}$

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Examples:

- $\{1, 2, 3\}$ is **finer** than $\{1, 3\}$
- $\{1, 1.1, 2, 2.5, 3\}$ is **coarser** than $\{1, 1.1, 1.5, 2, 2.5, 2.9, 3\}$
- $\{1, 1.5, 2, 2.5, 3\}$ and $\{1, 1.2, 1.4, 1.8, 2, 2.2, 2.4, 2.8, 3\}$ are **incomparable**

Riemann integrability

Definition

We write

$$\int_a^b f(x)dx = A$$

if for all $\epsilon > 0$, there exists a partition P_ϵ such that for any finer partition P

$$|R(P, f, \{t_k\}) - A| < \epsilon$$

for all choices of sample points $\{t_k\}$ of P .

In this case we say f is **Riemann integrable** on $[a, b]$ and call the value A the **Riemann integral**.

A first example

Problem

Let

$$f(x) = \begin{cases} 1, & x < 2 \\ 2, & x \geq 2 \end{cases}$$

What is $\int_1^3 f(x)dx$?

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What is $\int_1^3 f(x)dx$?

- Calculus experience says: $\int_1^3 f(x)dx = 3$.

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Let

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What is $\int_1^3 f(x)dx$?

- Calculus experience says: $\int_1^3 f(x)dx = 3$.
- How do we prove it?

INTUITION: the "action" is near $x = 2$, so we should try to make the partition denser there

A first example

Let $\epsilon > 0$.

A first example

Let $\epsilon > 0$. (WLOG $\epsilon < 1$)

Choose $P_\epsilon = \{1, 2 - \epsilon/4, 2, 3\}$

A refinement P of P_ϵ will look like

$$P = \{x_0 = 1, x_1, \dots, x_\ell = 2 - \epsilon, x_{\ell+1}, \dots, x_m = 2, x_{m+1}, \dots, x_n = 3\}$$

Then for any sample points, the Riemann sum is

$$\begin{aligned} R(P, f, \{t_k\}) &= \sum_{k=1}^n f(t_k) \Delta x_k \\ &= \sum_{k=1}^{\ell} f(t_k) \Delta x_k + \sum_{k=\ell+1}^m f(t_k) \Delta x_k + \sum_{k=m+1}^n f(t_k) \Delta x_k \end{aligned}$$

A first example

$$\begin{aligned}
 &= \sum_{k=1}^{\ell} f(t_k) \Delta x_k + \sum_{k=\ell+1}^{m-1} f(t_k) \Delta x_k + f(t_m) \Delta x_m + \sum_{k=m+1}^n f(t_k) \Delta x_k \\
 &= \sum_{k=1}^{\ell} 1 \Delta x_k + \sum_{k=\ell+1}^{m-1} 1 \Delta x_k + f(t_m) \Delta x_m + \sum_{k=m+1}^n 2 \Delta x_k \\
 &= (x_{\ell} - x_0) + (x_{m-1} - x_{\ell}) + f(t_m)(x_m - x_{m-1}) + 2(x_n - x_m) \\
 &= 1 - \epsilon/4 + (x_{m-1} - x_{\ell}) + f(t_m)(x_m - x_{m-1}) + 2
 \end{aligned}$$

Therefore

$$\begin{aligned}
 |R(P, f, \{t_k\}) - 3| &= | -\epsilon/4 + (x_{m-1} - x_{\ell}) + f(t_m)(x_m - x_{m-1}) | \\
 &< \epsilon/4 + \epsilon/4 + 2\epsilon/4 = \epsilon
 \end{aligned}$$

Riemann-Stieltjes sums

Notation: given a real-valued function α on $[a, b]$

$$\Delta\alpha_k(P) = \alpha(x_k) - \alpha(x_{k-1}), \quad k = 1, 2, \dots, n.$$

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Definition

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$.

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Definition

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. A sum of the form

$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^n f(t_k) \Delta\alpha_k(P),$$

for some set of sample points $\{t_k\}$ of P .

Riemann-Stieltjes integral

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Let f and α be real-valued functions on $[a, b]$. We say f is **Riemann integrable** with respect to α on $[a, b]$ if there exists $A \in \mathbb{R}$ with the following property:

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Let f and α be real-valued functions on $[a, b]$. We say f is **Riemann integrable** with respect to α on $[a, b]$ if there exists $A \in \mathbb{R}$ with the following property: for all $\epsilon > 0$, there exists a partition P_ϵ of $[a, b]$ such that for all refinements P of P_ϵ

$$|S(P, f, \alpha, \{t_k\}) - A| < \epsilon,$$

for all sets of sample points $\{t_k\}$ of P .

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$$|S(P, f, \alpha, \{t_k\}) - A| < \epsilon,$$

for all sets of sample points $\{t_k\}$ of P .

Notation: in this case A is denoted by $\int_a^b f d\alpha$ or $\int_a^b f(x) d\alpha(x)$

Challenge!

Problem

Suppose that a, b are real numbers with $a < b$ and let α be a real-valued function on $[a, b]$.

Prove that a constant function $f(x) = c$ is integrable with respect to α on $[a, b]$ and

$$\int_a^b f d\alpha = c\alpha(b) - c\alpha(a).$$

Challenge!

Problem

Suppose that a, b are real numbers with $a < b$ and let α be a constant function on $a < b$.

If $f(x)$ is a function on $[a, b]$, show that $f(x)$ is integrable with respect to α on $[a, b]$.

What is $\int_a^b f d\alpha$?

Challenge!

Problem

Suppose that a, b, c are real numbers with $a < c < b$ and let α be the step function

$$\alpha(x) = \begin{cases} 0, & x < c \\ 1, & x \geq c \end{cases}$$

Show that if $f(x)$ is a real function on $[a, b]$ which is continuous at c , then f is integrable with respect to α on $[a, b]$.

What is $\int_a^b f d\alpha$?

Linearity of integration

Theorem

Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.

Linearity of integration

Theorem

Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.

Then for any $c_1, c_2 \in \mathbb{R}$, $c_1 f + c_2 g$ is Riemann integrable with respect to α and

$$\int_a^b (c_1 f + c_2 g) d\alpha = c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha.$$

Linearity of integration

Proof.

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Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.

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Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.

Choose a partition P'_ϵ such that

$$|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| < \epsilon/2|c_1|,$$

for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P .

Linearity of integration

Proof.

Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.

Choose a partition P'_ϵ such that

$$|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| < \epsilon/2|c_1|,$$

for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P .
Also, choose a partition P''_ϵ such that

$$|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| < \epsilon/2|c_2|,$$

for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P .

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Proof.

Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.

Choose a partition P'_ϵ such that

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for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P .
Let $P_\epsilon = P'_\epsilon \cup P''_\epsilon$.

Linearity of integration

Proof.

Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.

Choose a partition P'_ϵ such that

$$|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| < \epsilon/2|c_1|,$$

for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P . Also, choose a partition P''_ϵ such that

$$|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| < \epsilon/2|c_2|,$$

for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P . Let $P_\epsilon = P'_\epsilon \cup P''_\epsilon$. Then any refinement P of P_ϵ , is a refinement of P'_ϵ and P''_ϵ . □

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Proof.

Therefore if P is a refinement of P_ϵ , and $\{t_k\}$ is a set of sample points of P , then by the triangle inequality

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Proof.

Therefore if P is a refinement of P_ϵ , and $\{t_k\}$ is a set of sample points of P , then by the triangle inequality

$$\begin{aligned} & |S(P, c_1 f + c_2 g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)| \\ &= |c_1 S(P, f, \alpha, \{t_k\}) + c_2 S(P, g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)| \\ &= |c_1 S(P, f, \alpha, \{t_k\}) - c_1 \int_a^b f d\alpha + c_2 S(P, g, \alpha, \{t_k\}) - c_2 \int_a^b g d\alpha| \\ &\leq |c_1| \cdot |S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| + |c_2| \cdot |S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| \\ &\qquad\qquad\qquad < \epsilon/2 + \epsilon/2 = \epsilon. \end{aligned}$$

Linearity of integration

Proof.

Therefore if P is a refinement of P_ϵ , and $\{t_k\}$ is a set of sample points of P , then by the triangle inequality

$$\begin{aligned}
 & |S(P, c_1 f + c_2 g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)| \\
 &= |c_1 S(P, f, \alpha, \{t_k\}) + c_2 S(P, g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)| \\
 &= |c_1 S(P, f, \alpha, \{t_k\}) - c_1 \int_a^b f d\alpha + c_2 S(P, g, \alpha, \{t_k\}) - c_2 \int_a^b g d\alpha| \\
 &\leq |c_1| \cdot |S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| + |c_2| \cdot |S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| \\
 &\qquad\qquad\qquad < \epsilon/2 + \epsilon/2 = \epsilon.
 \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, this proves the theorem. □ ↻ 🔍

Linearity of integration

Theorem

Suppose that f is Riemann integrable with respect to both α and β on $[a, b]$.

Linearity of integration

Theorem

Suppose that f is Riemann integrable with respect to both α and β on $[a, b]$.

Then for any $c_1, c_2 \in \mathbb{R}$, f is Riemann integrable with respect to $c_1\alpha + c_2\beta$ and

$$\int_a^b f d(c_1\alpha + c_2\beta) = c_1 \int_a^b f d\alpha + c_2 \int_a^b f d\beta.$$

Challenge!

Problem

Prove the last theorem!

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$.

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$. If $\int_a^c f d\alpha$ and $\int_c^b f d\alpha$ exist, then $\int_a^b f d\alpha$ exists and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$. If $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist, then $\int_a^b f d\alpha$ exists and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Conversely, if $\int_a^b f d\alpha$ exists then $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Integration by parts

The integrand and integrator play dual roles:

Theorem

Integration by parts Let f be integrable with respect to α on $[a, b]$.

Then α is integrable with respect to f on $[a, b]$ and

$$\int_a^b f d\alpha = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha df.$$

Relation to Riemann integrals

Theorem

Let f be integrable with respect to α on $[a, b]$.

If α is continuous on $[a, b]$ and continuously differentiable on (a, b) , then $f(x)\alpha'(x)$ is integrable with respect to x on $[a, b]$ and

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x) dx.$$